LLMs from First Principles: Sources

Best existing material per module. **Primary** = read as the main text; **Companion** = read alongside our PDF. All free unless marked (paid).

Module	Resource	Link	Role
M0 Math	Deisenroth, Faisal, Ong, *Mathematics for Machine Learning*	https://mml-book.github.io/	Primary
	Goodfellow, Bengio, Courville, *Deep Learning* (ch. 2 to 8)	https://www.deeplearningbook.org/	Primary
	MacKay, *Information Theory, Inference, and Learning Algorithms*	author's site (free PDF)	Companion
	Goldberg, "What Every Computer Scientist Should Know About Floating-Point Arithmetic" (1991)	search title	Companion
M1 History	Jurafsky & Martin, *Speech and Language Processing*, 3rd ed. draft	https://web.stanford.edu/~jurafsky/slp3/	Companion
	Our Chapter 1 PDF	(done)	Primary
M2 Transformer	Karpathy, *Neural Networks: Zero to Hero* + nanoGPT	https://karpathy.ai/zero-to-hero.html	Primary
	Rush et al., *The Annotated Transformer*	https://nlp.seas.harvard.edu/annotated-transformer/	Primary
	Prince, *Understanding Deep Learning* (ch. 12)	https://udlbook.github.io/udlbook/	Companion
M2b Training	Stanford CS336, *Language Modeling from Scratch* (lectures + 5 assignments)	http://cs336.stanford.edu/	Primary
M3 GPUs	Horace He, "Making Deep Learning Go Brrrr From First Principles"	https://horace.io/brrr_intro.html	Primary
	Simon Boehm, "How to Optimize a CUDA Matmul Kernel"	https://siboehm.com/articles/22/CUDA-MMM	Primary
	Hwu, Kirk, El Hajj, *Programming Massively Parallel Processors* (paid)	book	Companion
M3–M5	Google, *How to Scale Your Model* (rooflines, sharding, training, inference, serving, GPUs)	https://jax-ml.github.io/scaling-book/	Primary
M5 Parallel	Hugging Face, *The Ultra-Scale Playbook*	https://huggingface.co/spaces/nanotron/ultrascale-playbook	Primary
	Stas Bekman, *Machine Learning Engineering* (hardware, network, training, inference)	https://github.com/stas00/ml-engineering	Companion
M4 Serving	Papers: Orca, vLLM/PagedAttention, Sarathi-Serve, DistServe, speculative decoding	arXiv / OSDI / SOSP	Primary
	How to Scale Your Model, parts 7 to 8	see above	Companion
M6 MoE	Fedus, Dean, Zoph, "A Review of Sparse Expert Models in Deep Learning" (2022)	https://arxiv.org/abs/2209.01667	Companion
	Papers: Shazeer 2017, GShard, Switch, ST-MoE, DeepSeekMoE, DeepSeek-V3	arXiv	Primary
M7 Fabric	Barroso, Hölzle, Ranganathan, *The Datacenter as a Computer*	publisher	Companion
	Papers: Clos 1953, Al-Fares 2008, Jupiter Rising/Evolving, TPU v4, DCQCN, RotorNet, Sirius	SIGCOMM / ISCA	Primary
	Dally & Towles, *Principles and Practices of Interconnection Networks* (paid)	book	Companion
M8 Capstone	*How to Scale Your Model* + model technical reports	see above	Companion

Where we write full chapters (no good single source exists)

M0	LLM-focused math: floating point and FP8, online softmax, AdamW, signal propagation.
M4 Serving	Batching, paged KV, disaggregation and the speculative-decoding proof as one derived story.
M6 MoE	Through DeepSeek-V3, with all-to-all bytes per token and expert-parallel serving.
M7 Fabric & optics	Optical circuit switching for ML traffic; reconfiguration time vs burst length.
M8 Capstone	Compute, memory, network, power and dollars per token in one model.

Links checked 11 Oct 2026, except MacKay (site did not respond; the book is free on the author's page) and the two textbooks marked paid.